

Top 25 DSA Interview Questions

1. What is Time Complexity and Space Complexity?
2. Explain Big O, Big Omega, and Big Theta.
3. Array vs Linked List.
4. Implement Stack and Queue.
5. What is a Hash Table? Handle collisions.
6. Binary Search and its complexity.
7. Merge Sort vs Quick Sort.
8. When would you use Heap?
9. What is a Binary Search Tree?
10. AVL Tree vs Red-Black Tree.
11. Explain DFS vs BFS.
12. Detect a cycle in a graph.
13. Topological Sorting.
14. Dijkstra vs Bellman-Ford.
15. Minimum Spanning Tree (Prim/Kruskal).
16. Trie and its applications.
17. Sliding Window technique.
18. Two Pointer technique.
19. Recursion vs Iteration.
20. Dynamic Programming: Memoization vs Tabulation.
21. Longest Common Subsequence.
22. 0/1 Knapsack.
23. Kadane's Algorithm.
24. Backtracking examples.
25. How would you optimize an $O(n^2)$ solution?